UNIVERSITY NAME

Analysis IV — Final Exam

Year: 2024

Exercise 1:

Determine the nature (convergence/divergence) of the following numerical series of general term:

$$a)\ u_n = \frac{n^n}{(2n)!}$$
$$b)\ u_n = \frac{n + \cos n}{e^n + \sin n}$$
$$c)\ u_n = \sqrt{1 + \frac{(-1)^n}{\sqrt{n}}} - 1$$

Answer Area

a) Use Stirling's formula: $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$

Then:

$$(2n)! \sim \sqrt{4\pi n}\left(\frac{2n}{e}\right)^{2n} = 2\sqrt{\pi n} \cdot \frac{(2n)^{2n}}{e^{2n}}$$

So:

$$u_n \sim \frac{n^n}{2\sqrt{\pi n} \cdot \frac{(2n)^{2n}}{e^{2n}}} = \frac{e^{2n}}{2\sqrt{\pi n}} \cdot \frac{n^n}{(2n)^{2n}} = \frac{e^{2n}}{2\sqrt{\pi n}} \cdot \frac{1}{2^{2n} n^n} = \frac{1}{2\sqrt{\pi n}}\left(\frac{e^2}{4n}\right)^n$$

As $n \to \infty$, $\left(\frac{e^2}{4n}\right)^n \to 0$ very fast (exponential decay).

More precisely, use the ratio test:

$$\frac{u_{n+1}}{u_n} = \frac{(n+1)^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{n^n} = \frac{(n+1)^{n+1}}{n^n} \cdot \frac{(2n)!}{(2n+2)!}$$

$$= \left(1 + \frac{1}{n}\right)^n \cdot (n+1) \cdot \frac{1}{(2n+1)(2n+2)} \sim e \cdot n \cdot \frac{1}{4n^2} = \frac{e}{4n} \to 0$$

Since $\frac{u_{n+1}}{u_n} \to 0$, the series $\sum u_n$ converges by the ratio test.

b) Compare with geometric series:

$$0 \le u_n = \frac{n + \cos n}{e^n + \sin n} \le \frac{n+1}{e^n - 1} \quad (\text{since } \sin n \ge -1)$$

For large n, e^n dominates, so:

$$u_n \sim \frac{n}{e^n}$$

Since $\sum \frac{n}{e^n}$ converges (by ratio test: $\frac{n+1}{e^{n+1}} \cdot \frac{e^n}{n} = \frac{n+1}{ne} \to \frac{1}{e} < 1$), by comparison, $\sum u_n$ converges.

c) Use the expansion $\sqrt{1+\epsilon} - 1 = \frac{\epsilon}{2} - \frac{\epsilon^2}{8} + o(\epsilon^2)$:

Let $\epsilon = \frac{(-1)^n}{\sqrt{n}}$, then:

$$u_n = \frac{(-1)^n}{2\sqrt{n}} - \frac{1}{8n} + o\left(\frac{1}{n}\right)$$

The series $\sum \frac{(-1)^n}{2\sqrt{n}}$ converges by the alternating series test (Leibniz).

The series $\sum -\frac{1}{8n}$ diverges (harmonic series).

Since the divergent part $-\frac{1}{8n}$ dominates, the series $\sum u_n$ diverges.

Exercise 2:

Consider the series of functions $\sum b_n$ defined by:

$$f_n(x) = \frac{1}{n + n^2 x}$$

We denote:

$$f(x) = \sum_{n=1}^{\infty} f_n(x)$$

1. Determine the domain of definition of f.
2. Show that f is continuous on $\mathbb{R}_+^*$ and compute:

   $$\lim_{x\to\infty} f(x); \lim_{x\to 0} f(x)$$

   deduction.
3. Show that f is of class C^1 on $\mathbb{R}_+^*$ and give its derivative.
4. Sketch the variation table of f on $\mathbb{R}_+^*$ and draw its graph.
5. (a) Show the existence and compute the integral:

   $$\int_1^{\infty} \frac{1}{y + y^2 x}\, dy$$

   (b) Show that:

   $$\log \frac{x+1}{x} \le f(x) \le \log\left(\frac{1+2x}{2x}\right) + \frac{1}{1+x}$$

   Deduce an equivalent of $f(x)$ at 0.

Answer Area

1. Domain of definition:

   For $x > 0$: $f_n(x) = \frac{1}{n(1+nx)} \sim \frac{1}{n^2 x}$ as $n \to \infty$.

   Since $\sum \frac{1}{n^2}$ converges, $\sum f_n(x)$ converges for all $x > 0$.

   For $x = 0$: $f_n(0) = \frac{1}{n}$, and $\sum \frac{1}{n}$ diverges.

   For $x < 0$: There might be problems when denominator vanishes. For $x = -\frac{1}{k}$ with $k \in \mathbb{N}^*$, $f_k(x)$ is undefined.

2. Continuity and limits:

   For any $\delta > 0$, on $[\delta, \infty)$:

   $$|f_n(x)| = \frac{1}{n + n^2 x} \leq \frac{1}{n^2 \delta}$$

   Since $\sum \frac{1}{n^2 \delta}$ converges, the series converges uniformly on $[\delta, \infty)$. Each f_n is continuous, so f is continuous on $[\delta, \infty)$ for any $\delta > 0$, hence on $(0, \infty)$.

   Limits:

   As $x \to \infty$: For fixed n, $\lim_{x\to\infty} f_n(x) = 0$. By uniform convergence, can interchange limit and sum:

   $$\lim_{x\to\infty} f(x) = \sum_{n=1}^{\infty} \lim_{x\to\infty} f_n(x) = 0$$

   As $x \to 0^+$: $f_n(x) \sim \frac{1}{n}$ for small x, so the series behaves like the harmonic series.

   More rigorously, for any N:

   $$f(x) \geq \sum_{n=1}^{N} f_n(x) = \sum_{n=1}^{N} \frac{1}{n + n^2 x}$$

   So:

   $$\liminf_{x\to 0^+} f(x) \geq \sum_{n=1}^{N} \lim_{x\to 0^+} f_n(x) = \sum_{n=1}^{N} \frac{1}{n}$$

   Since this holds for all N and $\sum_{n=1}^{N} \frac{1}{n} \to \infty$ as $N \to \infty$, we have $\lim_{x\to 0^+} f(x) = +\infty$.

3. Differentiability:

   Compute derivative:

   $$f_n'(x) = -\frac{n^2}{(n + n^2 x)^2} = -\frac{1}{(1 + nx)^2}$$

On $[\delta, \infty)$ with $\delta > 0$:

$$|f_n'(x)| = \frac{1}{(1+nx)^2} \leq \frac{1}{(1+n\delta)^2} \leq \frac{1}{n^2\delta^2}$$

Since $\sum \frac{1}{n^2\delta^2}$ converges, $\sum f_n'$ converges uniformly on $[\delta, \infty)$. Therefore f is C^1 on $(0, \infty)$ and:

$$f'(x) = \sum_{n=1}^{\infty} f_n'(x) = -\sum_{n=1}^{\infty} \frac{1}{(1+nx)^2}$$

4. Variation table and graph:

   Since $f'(x) < 0$ for all $x > 0$, f is strictly decreasing.

   Limits:

   - $f(0^+) = +\infty$
   - $f(+\infty) = 0$

   So f decreases from $+\infty$ to 0.

   Graph: Hyperbola-like shape, decreasing, asymptote $y = 0$ at infinity, vertical asymptote at $x = 0$.

5. (a) Compute the integral:

   $$\int_1^{\infty} \frac{1}{y + y^2 x} dy = \int_1^{\infty} \frac{1}{y(1+xy)} dy$$

   Use partial fractions: $\frac{1}{y(1+xy)} = \frac{1}{y} - \frac{x}{1+xy}$

   So:

   $$\int_1^{\infty} \frac{1}{y(1+xy)} dy = \lim_{R\to\infty} [\ln y - \ln(1+xy)]_1^R$$

   $$= \lim_{R\to\infty} \ln\left(\frac{1}{x + 1/R}\right) + \ln(1+x) = -\ln x + \ln(1+x) = \ln\left(\frac{1+x}{x}\right)$$

   (b) Compare sum with integral:

   Since $f_n(x)$ is decreasing in n for fixed x, we have:

   $$\int_1^{\infty} \frac{1}{y+y^2x} dy \leq \sum_{n=1}^{\infty} \frac{1}{n+n^2x} \leq f_1(x) + \int_1^{\infty} \frac{1}{y+y^2x} dy$$

   That is:

   $$\ln\left(\frac{1+x}{x}\right) \leq f(x) \leq \frac{1}{1+x} + \ln\left(\frac{1+x}{x}\right)$$

   Wait, the problem gives: $f(x) \leq \log\left(\frac{1+2x}{2x}\right) + \frac{1}{1+x}$

   There's a slight difference. Possibly using $\sum_{n=1}^{\infty} \leq \int_0^{\infty}$ instead of $\int_1^{\infty}$.

   At $x \to 0^+$: $\ln\left(\frac{1+x}{x}\right) \sim -\ln x$

   So from the lower bound $f(x) \geq \ln\left(\frac{1+x}{x}\right) \sim -\ln x$, and the upper bound $f(x) \leq \text{constant} + \ln\left(\frac{1+x}{x}\right) \sim -\ln x$, we get:

   $$f(x) \sim -\ln x \quad \text{as } x \to 0^+$$

Exercise 3:

Let $f : x \in (-1, 1) \longrightarrow \frac{\arcsin x}{\sqrt{1-x^2}}$.

1. Justify that f can be expanded as a power series.
2. Show that f satisfies a first-order differential equation.
3. Deduce the power series expansion of f.
4. Deduce that for any integer n:

$$\sum_{k=0}^{n} \frac{1}{2k+1} \binom{2k}{k} \binom{2(n-k)}{n-k} = \frac{2^{4n}(n!)^2}{(2n+1)!}$$

It is recalled that:

$$\frac{1}{\sqrt{1-x^2}} = \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n}(n!)^2} x^{2n}$$

Answer Area

1. Power series expansion:

   Both $\arcsin x$ and $\frac{1}{\sqrt{1-x^2}}$ are analytic on $(-1, 1)$:

   - $\arcsin x = \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n}(n!)^2} \frac{x^{2n+1}}{2n+1}$
   - $\frac{1}{\sqrt{1-x^2}} = \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n}(n!)^2} x^{2n}$

   Their product is also analytic on $(-1, 1)$, so f can be expanded as a power series.

2. Differential equation:

   Compute derivative:

$$f'(x) = \frac{\frac{1}{\sqrt{1-x^2}} \cdot \sqrt{1-x^2} - \arcsin x \cdot \frac{-x}{\sqrt{1-x^2}}}{1-x^2}$$

$$= \frac{1 + \frac{x \arcsin x}{\sqrt{1-x^2}}}{1-x^2} = \frac{1 + x f(x)}{1-x^2}$$

   So f satisfies:

$$f'(x) = \frac{1 + x f(x)}{1-x^2}$$

3. Power series expansion:

   Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$.

   From the differential equation: $(1-x^2)f'(x) - x f(x) = 1$

Compute:

$$f'(x) = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$$
$$-x f(x) = -\sum_{n=0}^{\infty} a_n x^{n+1} = -\sum_{n=1}^{\infty} a_{n-1} x^n$$

So the LHS becomes:

$$\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n - \sum_{n=2}^{\infty} (n-1) a_{n-1} x^n - \sum_{n=1}^{\infty} a_{n-1} x^n = 1$$

Write terms by degree:

- Degree 0: $a_1 = 1$
- Degree 1: $2a_2 - a_0 = 0 \Rightarrow a_2 = \frac{a_0}{2}$
- Degree $n \geq 2$: $(n+1)a_{n+1} - (n-1)a_{n-1} - a_{n-1} = 0$

$$\Rightarrow (n+1)a_{n+1} = n a_{n-1} \Rightarrow a_{n+1} = \frac{n}{n+1} a_{n-1}$$

Since $f(0) = 0$ (as $\arcsin 0 = 0$), we have $a_0 = 0$.

Then from recurrence:

$$\begin{aligned}
a_0 &= 0 \\
a_1 &= 1 \\
a_2 &= \frac{a_0}{2} = 0 \\
a_3 &= \frac{2}{3} a_1 = \frac{2}{3} \\
a_4 &= \frac{3}{4} a_2 = 0 \\
a_5 &= \frac{4}{5} a_3 = \frac{4}{5} \cdot \frac{2}{3} = \frac{8}{15}
\end{aligned}$$

So only odd coefficients are non-zero:

$$a_{2n+1} = \frac{2n}{2n+1} a_{2n-1} = \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdots \frac{2}{3} \cdot 1$$
$$= \frac{(2n)!!}{(2n+1)!!} = \frac{2^n n!}{(2n+1)!!}$$

Alternatively: $a_{2n+1} = \frac{2^{2n}(n!)^2}{(2n+1)!}$

So:

$$f(x) = \sum_{n=0}^{\infty} \frac{2^{2n}(n!)^2}{(2n+1)!} x^{2n+1}$$

4. Combinatorial identity:

From $f(x) = \frac{\arcsin x}{\sqrt{1-x^2}} = \left(\sum_{k=0}^{\infty} \frac{(2k)!}{2^{2k}(k!)^2} \frac{x^{2k+1}}{2k+1}\right) \left(\sum_{j=0}^{\infty} \frac{(2j)!}{2^{2j}(j!)^2} x^{2j}\right)$

The coefficient of x^{2n+1} is:

$$\sum_{k=0}^{n} \frac{(2k)!}{2^{2k}(k!)^2} \frac{1}{2k+1} \cdot \frac{(2(n-k))!}{2^{2(n-k)}((n-k)!)^2}$$

But from part (3), this coefficient is also $\frac{2^{2n}(n!)^2}{(2n+1)!}$.

So:

$$\sum_{k=0}^{n} \frac{1}{2k+1} \cdot \frac{(2k)!}{2^{2k}(k!)^2} \cdot \frac{(2(n-k))!}{2^{2(n-k)}((n-k)!)^2} = \frac{2^{2n}(n!)^2}{(2n+1)!}$$

Multiply both sides by 2^{2n}:

$$\sum_{k=0}^{n} \frac{1}{2k+1} \binom{2k}{k} \binom{2(n-k)}{n-k} = \frac{2^{4n}(n!)^2}{(2n+1)!}$$